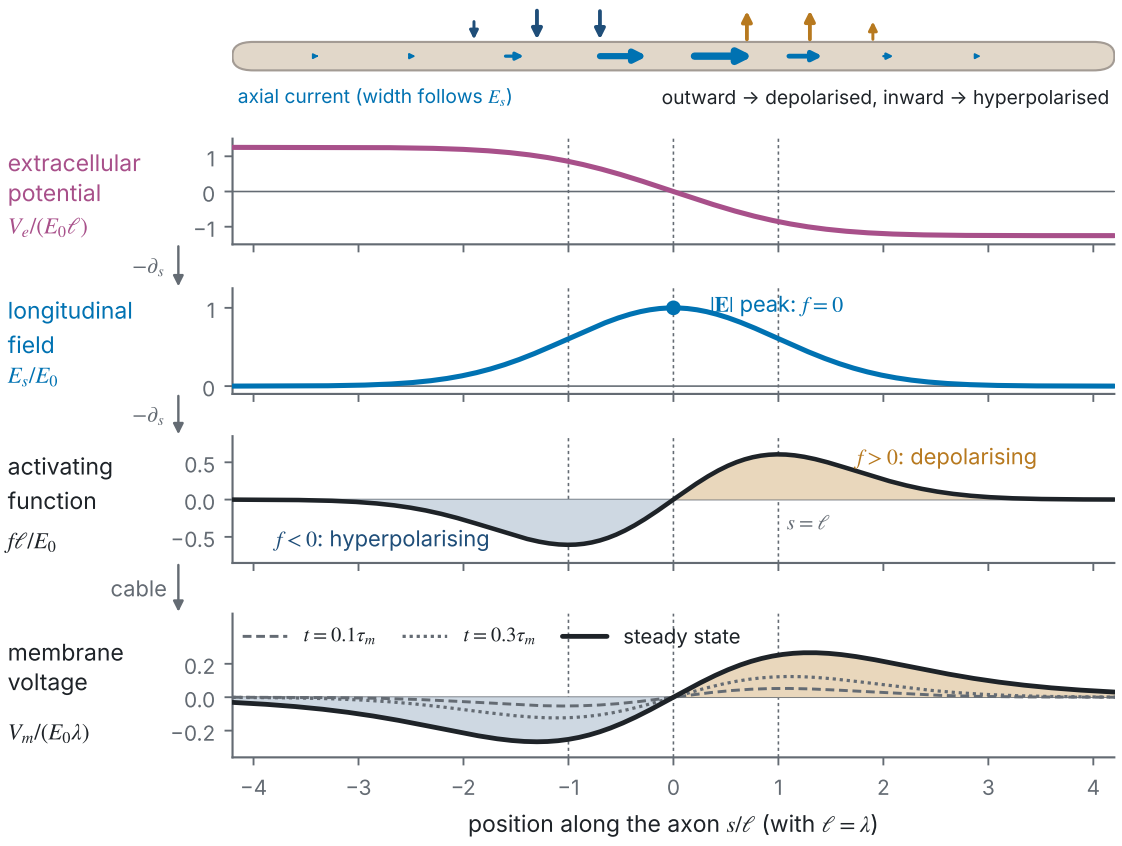

axial current (width follows E_s)
outward → depolarised, inward → hyperpolarised
extracellular potential
$V_e/(E_0\ell)$
1
0
−1
$-\partial_s$
longitudinal field
E_s/E_0
|E| peak: $f=0$
1
0
$-\partial_s$
activating function
$f\ell/E_0$
0.5
0.0
−0.5
$f>0$: depolarising
$f<0$: hyperpolarising
$s=\ell$
cable
membrane voltage
$V_m/(E_0\lambda)$
$t=0.1\tau_m$
$t=0.3\tau_m$
steady state
0.2
0.0
−0.2
−4
−3
−2
−1
0
1
2
3
4
position along the axon s/ℓ (with $\ell=\lambda$)